Lecture 12

Homogeneous spaces and compact induction

Objectives. *Let K be a compact Lie group and H a closed subgroup. Which irreducible representations of K occur in functions on K/H, and how often? We show that each irreducible representation V of K occurs $\dim V^H$ times. Induction from H extends this count to vector-valued functions.*

By the end of this lecture you will be able to:

- Construct the invariant integral on K/H and identify $L^2(K/H)$ with the right-H-invariant part of $L^2(K)$.
- Prove that each irreducible V occurs in $L^2(K/H)$ exactly $\dim V^H$ times.
- Define compact induction and prove Frobenius reciprocity with finite multiplicities.
- Decompose $L^2(S^2)$ under $\mathrm{SO}(3)$ and relate the summands to spherical harmonics.

Throughout, K is a compact Lie group, not necessarily connected, and $H \subseteq K$ is a closed subgroup. By Lecture 3, K/H is a compact manifold, and the projection $p\colon K \to K/H$, $p(x) = xH$, is continuous and open. The group K acts on K/H by $g \cdot xH = gxH$.

The Haar measures dx on K and dh on H have total mass 1. They are left, right and inversion invariant (Lecture 5). The space $L^2(K)$ is the completion of $C(K)$ for $\langle f_1, f_2\rangle = \int_K \overline{f_1}\, f_2\, dx$, and $C(K) \to L^2(K)$ is injective (Lecture 10).

We use the left and right translation representations

$$(\lambda(g)f)(x) = f(g^{-1}x), \qquad (\rho(g)f)(x) = f(xg).$$

These commuting representations are unitary and strongly continuous (Lecture 10).

12.1 Functions and integration on K/H

By the definition of the quotient topology, a function φ on K/H is continuous exactly when $\varphi \circ p$ is continuous. A right-H-invariant function on K is constant on cosets. Hence the pullback p^* is a linear bijection

$$p^*\colon C(K/H) \to C(K)^H := \{f \in C(K) : f(xh) = f(x) \text{ for all } x \in K,\ h \in H\}, \qquad p^*\varphi = \varphi \circ p. \tag{12.1}$$

Definition 12.1. The *invariant integral* on K/H is the linear functional

$$\int_{K/H} \varphi(\xi)\, d\xi := \int_K \varphi(p(x))\, dx, \qquad \varphi \in C(K/H).$$

Proposition 12.2. *(a) The integral is positive, $\int_{K/H} 1\, d\xi = 1$, and $\int_{K/H} \varphi(g^{-1}\xi)\, d\xi = \int_{K/H} \varphi(\xi)\, d\xi$ for $g \in K$.*

(b) For $f \in C(K)$, the H-average $f^H(x) = \int_H f(xh)\, dh$ lies in $C(K)^H$, and

$$\int_K f(x)\, dx = \int_K f^H(x)\, dx = \int_{K/H} \Big(\int_H f(xh)\, dh \Big)\, d(xH).$$

Proof. (a) Positivity and normalisation hold for dx. Invariance follows from $\varphi(g^{-1}p(x)) = \varphi(p(g^{-1}x))$ and left invariance of dx.

(b) The map $(x, h) \mapsto f(xh)$ is uniformly continuous on the compact space $K \times H$. So f^H is continuous. For $k \in H$, left invariance of dh gives $f^H(xk) = \int_H f(xkh)\, dh = f^H(x)$. Fubini for continuous functions on $K \times H$ and right invariance of dx give

$$\int_K f^H(x)\, dx = \int_H \int_K f(xh)\, dx\, dh = \int_H \int_K f(x)\, dx\, dh = \int_K f(x)\, dx.$$

By the pullback bijection (12.1), the last expression in (b) is $\int_K f^H\, dx$ written on K/H. ∎

Definition 12.3. The space $L^2(K/H)$ is the completion of $C(K/H)$ for $\langle \varphi, \psi \rangle = \int_{K/H} \overline{\varphi}\, \psi\, d\xi$. Also

$$L^2(K)^H = \{ f \in L^2(K) : \rho(h) f = f \text{ for all } h \in H \}.$$

The form is definite since $\int_{K/H} |\varphi|^2\, d\xi = \|\varphi \circ p\|^2$ and $C(K) \to L^2(K)$ is injective. The same identity shows that p^* is isometric. So p^* extends to an isometry $L^2(K/H) \to L^2(K)$. The subspace $L^2(K)^H$ is closed because each $\rho(h)$ is bounded.

Proposition 12.4. *There is a unique bounded operator A_H on $L^2(K)$ with $A_H f = f^H$ for $f \in C(K)$. It satisfies*

$$\langle f_1, A_H f_2 \rangle = \int_H \langle f_1, \rho(h) f_2 \rangle\, dh \qquad (f_1, f_2 \in L^2(K)). \tag{12.2}$$

It is the orthogonal projection onto $L^2(K)^H$. Consequently p^* maps $L^2(K/H)$ isometrically onto $L^2(K)^H$.

Proof. Let $f \in C(K)$. Cauchy–Schwarz on H gives $|f^H(x)|^2 \le (|f|^2)^H(x)$. Integrate and apply the averaging identity of Proposition 12.2(b) to $|f|^2$:

$$\|f^H\|^2 \le \int_K (|f|^2)^H\, dx = \int_K |f|^2\, dx = \|f\|^2.$$

Since $C(K)$ is dense, A_H exists uniquely, with $\|A_H\| \le 1$.

For continuous f_1, f_2, Fubini shows that both sides of (12.2) equal $\int_K \int_H \overline{f_1(x)} f_2(xh)\, dh\, dx$. Both sides are continuous in (f_1, f_2). On the right, the integrand is continuous in h by strong continuity and bounded by $\|f_1\|\, \|f_2\|$. So (12.2) holds on $L^2(K)$.

Now $\rho(h)^* = \rho(h^{-1})$ and dh is inversion invariant. So the weak formula (12.2) gives $\langle f_1, A_H f_2\rangle = \langle A_H f_1, f_2\rangle$, and A_H is self-adjoint.

Next, A_H fixes $L^2(K)^H$. If $f \in L^2(K)^H$, then the weak formula gives $\langle f_1, A_H f\rangle = \langle f_1, f\rangle$ for all f_1, so $A_H f = f$.

Finally, the range of A_H lies in $L^2(K)^H$. Let $k \in H$. For continuous f, $A_H f = f^H$ lies in $C(K)^H$ by the description of the H-average in Proposition 12.2(b), so $\rho(k)A_H f = A_H f$. Since $C(K)$ is dense and both sides are bounded in f, this identity holds for all f.

Hence A_H is a self-adjoint idempotent with range $L^2(K)^H$.

It remains to identify the image of p^*. Since p^* is an isometry, its image is the closure of $p^*(C(K/H)) = C(K)^H$. This closure lies in the closed subspace $L^2(K)^H$. Conversely, let $f \in L^2(K)^H$ and choose $f_n \in C(K)$ with $f_n \to f$. Then $f_n^H \in C(K)^H$ and $f_n^H = A_H f_n \to A_H f = f$. So the image is $L^2(K)^H$. ∎

Each $\lambda(g)$ commutes with every $\rho(h)$, so $\lambda(g)$ preserves $L^2(K)^H$. Transported by p^*, the operator $\lambda(g)$ acts on $C(K/H)$ by $\varphi \mapsto \varphi(g^{-1}\cdot)$. The resulting unitary representation of K on $L^2(K/H)$ is the *left regular representation*.

12.2 Multiplicities in $L^2(K/H)$

Let (π, V) be a finite-dimensional unitary representation and $u, v \in V$. The *matrix coefficient* is $c_{u,v}(x) = \langle u, \pi(x)v\rangle$. Unitarity gives

$$\lambda(g)c_{u,v} = c_{\pi(g)u,\,v}, \qquad \rho(g)c_{u,v} = c_{u,\,\pi(g)v}. \tag{12.3}$$

For the first identity, $c_{u,v}(g^{-1}x) = \langle u, \pi(g)^{-1}\pi(x)v\rangle = \langle \pi(g)u, \pi(x)v\rangle$.

The map $u \mapsto c_{u,v}$ is antilinear. The *conjugate representation* $\overline{V}$ is the set V with scalar multiplication $t \cdot u = \bar{t}u$, the same operators $\pi(g)$, and inner product $\langle u, u'\rangle_{\overline{V}} = \langle u', u\rangle$. It has the same invariant subspaces as V. So $\overline{V}$ is irreducible when V is, and $(\overline{V})^H = V^H$ as sets.

For each $\sigma \in \mathrm{Irr}(K)$ fix a unitary representative (π_σ, V_σ) of dimension d_σ. Write $\bar{\sigma}$ for the class of $\overline{V_\sigma}$, and let $E_\sigma = \mathrm{span}\{c_{u,v} : u, v \in V_\sigma\} \subseteq C(K)$. We use two earlier results:

(PW1) Schur orthogonality (Lecture 5): $\langle c_{u,v}, c_{u',v'}\rangle = d_\sigma^{-1}\langle u', u\rangle\langle v, v'\rangle$ for $u, v, u', v' \in V_\sigma$.

(PW2) Peter–Weyl (Lecture 10): the E_σ are mutually orthogonal and $L^2(K) = \widehat{\bigoplus}_\sigma E_\sigma$.

By Schur orthogonality (PW1), the linear map $\overline{V_\sigma} \otimes V_\sigma \to E_\sigma$, $u \otimes v \mapsto c_{u,v}$, is $d_\sigma^{-1/2}$ times an isometry. So it is bijective. By the coefficient identities (12.3), it carries $\bar{\pi}_\sigma \otimes 1$ to λ and $1 \otimes \pi_\sigma$ to ρ. Let Q_σ be the orthogonal projection onto E_σ. A unitary operator that preserves a finite-dimensional subspace also preserves its orthogonal complement. Hence Q_σ commutes with all $\lambda(g)$ and $\rho(g)$.

For $\sigma \in \mathrm{Irr}(K)$ let $E_\sigma^H = \mathrm{span}\{c_{u,v} : u \in V_\sigma,\ v \in V_\sigma^H\}$, a subspace of E_σ.

Proposition 12.5. *Let $P_\sigma = \int_H \pi_\sigma(h)\,dh$ be the orthogonal projection of V_σ onto V_σ^H (Lec-*

ture 5). Then $A_H c_{u,v} = c_{u,P_\sigma v}$. Moreover

$$L^2(K/H) \cong L^2(K)^H = \overline{\bigoplus_{\sigma \in \mathrm{Irr}(K)}} E_\sigma^H, \qquad E_\sigma^H \cong \overline{V_\sigma} \otimes V_\sigma^H,$$

where K acts on $\overline{V_\sigma} \otimes V_\sigma^H$ by $\bar\pi_\sigma \otimes 1$.

Proof. The finite-dimensional integral commutes with the functional $w \mapsto \langle u, \pi_\sigma(x) w\rangle$. So

$$(A_H c_{u,v})(x) = \int_H \langle u, \pi_\sigma(x)\pi_\sigma(h) v\rangle \, dh = \left\langle u, \pi_\sigma(x) \int_H \pi_\sigma(h) v \, dh \right\rangle = c_{u, P_\sigma v}(x).$$

Thus $A_H(E_\sigma) = E_\sigma^H$, and A_H fixes E_σ^H. Hence $E_\sigma^H \subseteq L^2(K)^H$. Conversely, let $f \in L^2(K)^H$. By Peter–Weyl (PW2), $f = \sum_\sigma Q_\sigma f$ in L^2. Since A_H is bounded,

$$f = A_H f = \sum_\sigma A_H Q_\sigma f, \qquad A_H Q_\sigma f \in E_\sigma^H.$$

Since E_σ^H lies in E_σ, the spaces E_σ^H are mutually orthogonal by Peter–Weyl (PW2). This proves the decomposition. The isomorphism with $\overline{V_\sigma} \otimes V_\sigma^H$ is the restriction of the bijection above. ∎

Induction in § 12.3 uses vector-valued functions and evaluates them at the identity e. An L^2 class has no value at a point. The next lemma shows that equivariant images of V_σ consist of continuous functions, so these evaluations make sense.

Let W be a finite-dimensional Hilbert space with orthonormal basis $w_1, \dots, w_n$. Then $L^2(K) \otimes W$ is the Hilbert space of functions $\sum_j f_j w_j$ with $f_j \in L^2(K)$. It contains $C(K, W) = C(K) \otimes W$. Let K act by $\lambda \otimes 1$.

Lemma 12.6. *Let $\sigma \in \mathrm{Irr}(K)$. Let $T : V_\sigma \to L^2(K) \otimes W$ be linear with $T\pi_\sigma(g) = (\lambda(g) \otimes 1)T$ for all $g \in K$. Then $T(V_\sigma) \subseteq E_{\bar\sigma} \otimes W$. In particular, every vector in $T(V_\sigma)$ is a continuous function.*

Proof. Fix $\eta \in \mathrm{Irr}(K)$. The operator $Q_\eta \otimes 1$ commutes with $\lambda(g) \otimes 1$. So $(Q_\eta \otimes 1)T$ is a K-map from V_σ into

$$E_\eta \otimes W \cong \overline{V_\eta} \otimes (V_\eta \otimes W),$$

where K acts only on the first factor. This space is a direct sum of $d_\eta \dim W$ copies of $\overline{V_\eta}$. By Schur's lemma, $(Q_\eta \otimes 1)T = 0$ unless $V_\sigma \cong \overline{V_\eta}$, that is, unless $\eta = \bar\sigma$. Apply Peter–Weyl (PW2) in each coordinate. For $v \in V_\sigma$,

$$Tv = \sum_\eta (Q_\eta \otimes 1) Tv = (Q_{\bar\sigma} \otimes 1) Tv \in E_{\bar\sigma} \otimes W.$$ ∎

Thus the values $(Tv)(x)$ are defined for all $v \in V_\sigma$ and $x \in K$.

Let (E, Π) be a unitary representation of K and $\sigma \in \mathrm{Irr}(K)$. The *multiplicity* of σ in E is $m_\sigma(E) = \dim \mathrm{Hom}_K(V_\sigma, E)$. Here Hom_K consists of the linear maps T with $T\pi_\sigma(g) = \Pi(g)T$. These maps are bounded because V_σ is finite-dimensional. The *σ-isotypic subspace* of E is the sum of the images of these maps.

Theorem 12.7. *For every $\sigma \in \operatorname{Irr}(K)$,*

$$m_\sigma\big(L^2(K/H)\big) = \dim V_\sigma^H.$$

The σ-isotypic subspace is $E_{\bar\sigma}^H \subseteq C(K/H)$, of dimension $d_\sigma \dim V_\sigma^H$. The space $L^2(K/H)$ is the Hilbert sum of its isotypic subspaces.

Proof. Identify $L^2(K/H)$ with $L^2(K)^H$ through p^*, as in Proposition 12.4. Let $T : V_\sigma \to L^2(K)^H$ be a K-map. By Lemma 12.6 on equivariant images, with $W = \mathbb{C}$, $T(V_\sigma) \subseteq E_{\bar\sigma}$. A vector of $E_{\bar\sigma}$ fixed by A_H lies in $A_H(E_{\bar\sigma}) = E_{\bar\sigma}^H$. Hence

$$\operatorname{Hom}_K\left(V_\sigma, L^2(K/H)\right) = \operatorname{Hom}_K\left(V_\sigma, E_{\bar\sigma}^H\right).$$

Since $V_{\bar\sigma} \cong \overline{V_\sigma}$, we have $\overline{V_{\bar\sigma}} \cong V_\sigma$. So Proposition 12.5 on the decomposition of $L^2(K)^H$ gives

$$E_{\bar\sigma}^H \cong \overline{V_{\bar\sigma}} \otimes V_{\bar\sigma}^H \cong V_\sigma \otimes V_{\bar\sigma}^H,$$

with K acting on the first factor only. By Schur's lemma,

$$\operatorname{Hom}_K\left(V_\sigma, V_\sigma \otimes V_{\bar\sigma}^H\right) \cong V_{\bar\sigma}^H.$$

A K-isomorphism $V_{\bar\sigma} \to \overline{V_\sigma}$ carries $V_{\bar\sigma}^H$ onto $(\overline{V_\sigma})^H$. This is the set V_σ^H with conjugated scalars, so its dimension is $\dim V_\sigma^H$. The images of the K-maps $V_\sigma \to E_{\bar\sigma}^H$ span $E_{\bar\sigma}^H \cong V_\sigma \otimes V_{\bar\sigma}^H$. Finally, $\sigma \mapsto \bar\sigma$ is a bijection of $\operatorname{Irr}(K)$, so the decomposition in Proposition 12.5 gives the Hilbert sum. ∎

Thus the summand E_σ^H of Proposition 12.5 is indexed by σ but is $\bar\sigma$-isotypic. The count is unchanged because $\dim V_\sigma^H = \dim V_{\bar\sigma}^H$. Exercise 12.1 shows this label shift for $K = \mathrm{U}(1)$.

12.3 Compact induction and Frobenius reciprocity

Let (τ, W) be a finite-dimensional unitary representation of H. Averaging over H gives every finite-dimensional representation of H an invariant inner product (Lecture 5).

Definition 12.8. The *induced representation* is

$$\operatorname{Ind}_H^K W = \{F \in C(K, W) : F(xh) = \tau(h)^{-1} F(x) \text{ for all } x \in K,\ h \in H\},$$

with K acting by $(\lambda(g)F)(x) = F(g^{-1}x)$.

Left translation preserves the covariance condition because left and right translations commute. For the trivial representation $W = \mathbb{C}$ of H, we recover $C(K)^H \cong C(K/H)$.

Theorem 12.9 (Frobenius reciprocity, continuous form)**.** *Let (π, V) be a finite-dimensional representation of K. The map*

$$\Phi :\ \operatorname{Hom}_K\left(V, \operatorname{Ind}_H^K W\right) \to \operatorname{Hom}_H(V|_H, W), \qquad \Phi(T)v = (Tv)(e),$$

is a linear bijection. Its inverse sends A to $\Psi(A)$, where $(\Psi(A)v)(x) = A\big(\pi(x)^{-1}v\big)$.

Proof. First, $\Phi(T)$ is H-equivariant. For $h \in H$, equivariance of T and covariance give

$$\Phi(T)\big(\pi(h)v\big) = (\lambda(h)Tv)(e) = (Tv)(h^{-1}) = \tau(h^{-1})^{-1}(Tv)(e) = \tau(h)\,\Phi(T)v.$$

Next, $\Psi(A)v$ is continuous because $x \mapsto \pi(x)^{-1}v$ is continuous. Since A is an H-map, $\Psi(A)v$ is covariant:

$$(\Psi(A)v)(xh) = A\big(\pi(h)^{-1}\pi(x)^{-1}v\big) = \tau(h)^{-1}A\big(\pi(x)^{-1}v\big).$$

The map $\Psi(A)$ is a K-map, since $\pi(x)^{-1}\pi(g) = \pi(g^{-1}x)^{-1}$ gives

$$(\Psi(A)\pi(g)v)(x) = (\Psi(A)v)(g^{-1}x) = (\lambda(g)\Psi(A)v)(x).$$

Finally, $\Phi(\Psi(A))v = A(v)$. In the other order,

$$(\Psi(\Phi(T))v)(x) = \big(T\pi(x^{-1})v\big)(e) = (\lambda(x^{-1})Tv)(e) = (Tv)(x). \qquad \blacksquare$$

Now pass to L^2. For $h \in H$ define $R(h) = \rho(h) \otimes \tau(h)$ on $L^2(K) \otimes W$. These operators are unitary and commute with $\lambda(g) \otimes 1$. For continuous F, $(R(h)F)(x) = \tau(h)F(xh)$. Set

$$L^2\text{-}\operatorname{Ind}_H^K W = \{F \in L^2(K) \otimes W : R(h)F = F \text{ for all } h \in H\}.$$

This closed K-invariant subspace is a unitary representation of K. For the trivial representation $W = \mathbb{C}$, this subspace is $L^2(K)^H$.

Theorem 12.10 (finite multiplicities). *For $\sigma \in \operatorname{Irr}(K)$,*

$$m_\sigma\big(L^2\text{-}\operatorname{Ind}_H^K W\big) = \dim \operatorname{Hom}_H(V_\sigma|_H, W) \le d_\sigma \dim W.$$

Proof. Let $T : V_\sigma \to L^2\text{-}\operatorname{Ind}_H^K W$ be a K-map. By Lemma 12.6 on equivariant images, each $F = Tv$ is continuous. Then $R(h)F$ and F are continuous and equal in L^2. Since $C(K) \otimes W \to L^2(K) \otimes W$ is injective, $\tau(h)F(xh) = F(x)$ for all x. So $F \in \operatorname{Ind}_H^K W$. Conversely, continuous covariant functions satisfy $R(h)F = F$. Hence

$$\operatorname{Hom}_K\big(V_\sigma, L^2\text{-}\operatorname{Ind}_H^K W\big) = \operatorname{Hom}_K\big(V_\sigma, \operatorname{Ind}_H^K W\big).$$

Frobenius reciprocity (Theorem 12.9) gives a linear bijection from this space onto $\operatorname{Hom}_H(V_\sigma|_H, W)$, so the dimensions agree. The bound holds because $\operatorname{Hom}_H(V_\sigma|_H, W)$ is a subspace of $\operatorname{Hom}(V_\sigma, W)$, which has dimension $d_\sigma \dim W$. $\blacksquare$

Remark 12.11 (duals and covariance). For $W = \mathbb{C}$ the count is $\dim \operatorname{Hom}_H(V_\sigma, \mathbb{C}) = \dim(V_\sigma^*)^H$. Since $\langle u, \pi_\sigma(h)v\rangle = \langle \pi_\sigma(h)^{-1}u, v\rangle$, the functional $\langle u, \cdot\rangle$ is H-invariant exactly when $u \in V_\sigma^H$. So $u \mapsto \langle u, \cdot\rangle$ is an antilinear bijection $V_\sigma^H \to (V_\sigma^*)^H$. Hence the count agrees with the multiplicity theorem (Theorem 12.7).

The inverse in $\tau(h)^{-1}$ makes the covariance condition compatible with products in H:

$$F(xh_1h_2) = \tau(h_2)^{-1}F(xh_1) = \tau(h_2)^{-1}\tau(h_1)^{-1}F(x) = \tau(h_1h_2)^{-1}F(x).$$

Remark 12.12 (scope). For compact K, we obtained a finite invariant integral. The irreducible representations were finite-dimensional, and Peter–Weyl applied. We proved a count of finite multiplicities, not a general adjunction for Hilbert representations. For noncompact G, the space G/H may carry no invariant measure. Lecture 17 treats that case.

12.4 The sphere $\mathrm{SO}(3)/\mathrm{SO}(2)$

Let $K = \mathrm{SO}(3)$, let e_1, e_2, e_3 be the standard basis of $\mathbb{R}^3$, and let

$$R_\theta = \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad H = \{R_\theta : \theta \in \mathbb{R}\} \cong \mathrm{SO}(2).$$

The subgroup H is the stabiliser of e_3. Every unit vector is the third column of some rotation, so the orbit map $q(x) = xe_3$ sends K onto S^2. By Lecture 3, q induces a K-equivariant diffeomorphism $K/H \to S^2$. So $F \mapsto F \circ q$ identifies $C(S^2)$ with $C(K)^H$, and $\lambda(g)(F \circ q) = (F \circ g^{-1}) \circ q$. We define $L^2(S^2)$ through this identification.

Let E be a finite-dimensional representation of H. An integer k is a *weight* of E if some nonzero $v \in E$ satisfies $R_\theta v = e^{ik\theta} v$ for all $\theta \in \mathbb{R}$. These vectors and 0 form the *weight-k space*. Suppose E has a basis (v_j) with $R_\theta v_j = e^{ik_j\theta} v_j$ for all θ. Then the weight-k space is spanned by the v_j with $k_j = k$. Indeed, a weight-k vector $\sum_j c_j v_j$ satisfies

$$c_j\big(e^{ik_j\theta} - e^{ik\theta}\big) = 0 \qquad \text{for all } \theta,$$

and this forces $c_j = 0$ when $k_j \neq k$.

From Lecture 8, the irreducibles of $\mathrm{SO}(3)$ are V_ℓ, $\ell = 0, 1, 2, \ldots$, of dimension $2\ell + 1$ and spin ℓ. In this section an unqualified V_ℓ means the $\mathrm{SO}(3)$-module. The $\mathrm{SU}(2)$-modules of Lecture 8 are always named as such. Choose the covering $\phi = \mathrm{Ad}$ in the ordered basis (X_1, X_2, X_3) of Lecture 4. The explicit calculation there gives

$$\phi\big(\mathrm{diag}(e^{i\theta/2}, e^{-i\theta/2})\big) = R_{-\theta}, \qquad \ker\phi = \{\pm I\}.$$

The pullback $\pi_\ell \circ \phi$ is the $\mathrm{SU}(2)$-module $V_{2\ell}$. The diagonal element in this display acts on a basis of $V_{2\ell}$ by the scalars $e^{i(\ell-j)\theta}$, $0 \le j \le 2\ell$. Replacing θ by $-\theta$ therefore gives

$$V_\ell \text{ has a basis } (v_k)_{-\ell \le k \le \ell} \text{ with } \pi_\ell(R_\theta) v_k = e^{ik\theta} v_k \text{ for all } \theta \in \mathbb{R}. \tag{12.4}$$

Theorem 12.13. *Let $\mathcal{Y}_\ell \subseteq L^2(S^2)$ be the V_ℓ-isotypic subspace. Then*

$$L^2(S^2) = \widehat{\bigoplus_{\ell \ge 0}} \mathcal{Y}_\ell, \qquad \mathcal{Y}_\ell \cong V_\ell, \qquad \mathcal{Y}_\ell \subseteq C(S^2).$$

Each integral-spin module occurs exactly once.

Proof. The space V_ℓ^H is the weight-zero space. By the weight basis (12.4) it is the line $\mathbb{C}v_0$, so $\dim V_\ell^H = 1$. Apply the multiplicity theorem (Theorem 12.7). ∎

Let $\mathcal{P}_\ell$ be the space of complex homogeneous polynomials of degree ℓ on $\mathbb{R}^3$, with $(\lambda(g)f)(v) = f(g^{-1}v)$. Let $\Delta = \partial_x^2 + \partial_y^2 + \partial_z^2$ and $\mathcal{H}_\ell = \{f \in \mathcal{P}_\ell : \Delta f = 0\}$. The chain rule gives $\Delta(f \circ g) = (\Delta f) \circ g$ for orthogonal g, so $\mathcal{H}_\ell$ is K-invariant. Restriction to S^2 is K-equivariant. Restriction is injective on $\mathcal{P}_\ell$, since $f(rv) = r^\ell f(v)$. Elements of $\mathcal{H}_\ell|_{S^2}$ are called *spherical harmonics* of degree ℓ.

Put $z_\pm = x \pm iy$. Since $R_\theta^{-1} = R_{-\theta}$,

$$(\lambda(R_\theta)z_+)(x,y,z) = (\cos\theta\, x + \sin\theta\, y) + i(-\sin\theta\, x + \cos\theta\, y) = e^{-i\theta} z_+.$$

Similarly $\lambda(R_\theta)z_- = e^{i\theta}z_-$ and $\lambda(R_\theta)z = z$. The monomials $z_+^a z_-^b z^c$ with $a+b+c = \ell$ form a basis of $\mathcal{P}_\ell$, and for all θ,

$$\lambda(R_\theta)\, z_+^a z_-^b z^c = e^{i(b-a)\theta}\, z_+^a z_-^b z^c, \qquad |b-a| \le \ell. \tag{12.5}$$

So every weight of $\mathcal{P}_\ell$ satisfies $|k| \le \ell$.

Proposition 12.14. *The space $\mathcal{Y}_\ell$ is spanned by the left translates of $f_\ell = z_+^\ell|_{S^2}$. Hence every element of $\mathcal{Y}_\ell$ is a spherical harmonic of degree ℓ.*

Proof. The polynomial z_+^ℓ is harmonic:

$$\partial_x^2 z_+^\ell + \partial_y^2 z_+^\ell = \ell(\ell-1)z_+^{\ell-2} + i^2\ell(\ell-1)z_+^{\ell-2} = 0.$$

Let U be the span of all $\lambda(g)f_\ell$. Then $U \subseteq \mathcal{H}_\ell|_{S^2}$ is finite-dimensional and K-invariant. Decompose U into irreducibles. A copy of V_j in U lies in the V_j-isotypic subspace $\mathcal{Y}_j$. Since $\mathcal{Y}_j \cong V_j$, this copy equals $\mathcal{Y}_j$. Thus $U = \bigoplus_{j\in S}\mathcal{Y}_j$ for a finite set S. Restriction is injective and equivariant, so the weights of U are weights of $\mathcal{P}_\ell$. Since V_j has weight j, this gives $S \subseteq \{0,\dots,\ell\}$.

Write $f_\ell = \sum_{j\in S} y_j$ with $y_j \in \mathcal{Y}_j$. Each $\lambda(R_\theta)$ preserves each $\mathcal{Y}_j$, and $\lambda(R_\theta)f_\ell = e^{-i\ell\theta}f_\ell$. Since the sum is direct, comparing components gives $\lambda(R_\theta)y_j = e^{-i\ell\theta}y_j$ for all θ. For $j < \ell$ the weight $-\ell$ does not occur in V_j, so $y_j = 0$. Hence $f_\ell \in \mathcal{Y}_\ell$ and $U \subseteq \mathcal{Y}_\ell$. Since $U \neq 0$ and $\mathcal{Y}_\ell$ is irreducible, $U = \mathcal{Y}_\ell$. ∎

Example 12.15 (degrees one and two)**.** We compute $\mathcal{Y}_1$ and $\mathcal{Y}_2$ and their H-fixed lines. Every linear polynomial is harmonic, so $\dim\mathcal{H}_1 = 3$. The map $\Delta\colon \mathcal{P}_2 \to \mathbb{C}$ is onto, since $\Delta(x^2) = 2$. So $\dim\mathcal{H}_2 = 6 - 1 = 5$.

By Proposition 12.14 on spherical harmonics, $\mathcal{Y}_\ell \subseteq \mathcal{H}_\ell|_{S^2}$. Restriction is injective, so $\mathcal{H}_\ell|_{S^2}$ has dimension $2\ell+1$ for $\ell = 1, 2$. By the sphere theorem (Theorem 12.13), also $\dim\mathcal{Y}_\ell = 2\ell+1$. So the inclusion is an equality for $\ell = 1, 2$:

$$\mathcal{Y}_1 = \operatorname{span}\{x,y,z\}|_{S^2}, \qquad \mathcal{Y}_2 = \operatorname{span}\{xy,\ yz,\ zx,\ x^2-y^2,\ 2z^2-x^2-y^2\}|_{S^2}.$$

By the monomial weights (12.5), the weight-zero space of $\mathcal{P}_2$ is spanned by $z_+z_- = x^2+y^2$ and z^2. Since $\Delta\big(a(x^2+y^2)+bz^2\big) = 4a+2b$, its harmonic line is spanned by $2z^2-x^2-y^2$. This function spans $\mathcal{Y}_2^H$, the line predicted by $\dim V_2^H = 1$. Similarly z spans $\mathcal{Y}_1^H$.

Exercise 12.5 proves $\mathcal{H}_\ell|_{S^2} = \mathcal{Y}_\ell$ for all ℓ. That exercise defines the Laplacian of the sphere and computes its eigenvalue on $\mathcal{H}_\ell$. Nothing in this lecture depends on it.

12.5 Self-checks

1. Let K be a finite group, regarded as a compact Lie group of dimension zero, and let $H \le K$ be a subgroup. What does the multiplicity theorem (Theorem 12.7) say?
   Answer. Here $L^2(K/H) = \mathbb{C}[K/H]$ is the permutation representation. An irreducible V occurs $\dim V^H$ times. This is finite-group Frobenius reciprocity for the trivial character of H.
2. Why not define $\Phi(T)v = (Tv)(e)$ directly for $T : V_\sigma \to L^2\text{-}\operatorname{Ind}_H^K W$?
   Answer. An L^2 class has no value at e. Lemma 12.6 on equivariant images first shows that T takes continuous values.
3. What are the multiplicities for $H = \{e\}$ and for $H = K$?
   Answer. For $H = \{e\}$, σ occurs d_σ times in $L^2(K)$. For $H = K$, a nontrivial irreducible has $V^K = 0$, so only the trivial representation occurs, once, as the constants.

Summary

The invariant integral on K/H is the Haar integral of pullbacks. The averaging operator A_H projects $L^2(K)$ onto $L^2(K)^H \cong L^2(K/H)$. Peter–Weyl then shows that each $\sigma \in \operatorname{Irr}(K)$ occurs in $L^2(K/H)$ exactly $\dim V_\sigma^H$ times.

Let V be a finite-dimensional representation of K and W a finite-dimensional unitary representation of H. Evaluation at e gives $\operatorname{Hom}_K(V, \operatorname{Ind}_H^K W) \cong \operatorname{Hom}_H(V|_H, W)$. Isotypic vectors in $L^2\text{-}\operatorname{Ind}_H^K W$ are continuous. So σ occurs in $L^2\text{-}\operatorname{Ind}_H^K W$ exactly $\dim \operatorname{Hom}_H(V_\sigma|_H, W)$ times.

On $S^2 = \mathrm{SO}(3)/\mathrm{SO}(2)$ each $\mathrm{SO}(3)$-module V_ℓ occurs once in $L^2(S^2)$, realised by spherical harmonics of degree ℓ.

Reading

- Bröcker and tom Dieck, *Representations of Compact Lie Groups* (1985), Section III.6: induction from closed subgroups and Frobenius reciprocity.
- Sepanski, *Compact Lie Groups* (2007), §2.3 and §§3.3–3.4: the assigned background route for this lecture and the sphere example.
- Knapp, *Lie Groups Beyond an Introduction*, digital second edition (2023), Section IX.2: induced representations of compact groups and Frobenius reciprocity.